

PAPER_952: 26-State HRes Spectral Ladder

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** spectral_ladder_26state.py (SpectralLadder26State) **Calculator:** SpectralLadder26StateCalc (CP4 #536) **CVW:** v2.0.0 compliant

Abstract

We derive the 26-state HRes (Hydrogen Resonance) spectral ladder, the energy progression across quantum layers from proto-H ($n = 1$) to proto-Fe ($n = 26$). The energy levels $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}$ form a geometrically growing ladder stabilizing the vacuum ratio $\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1$ and driving phonon resonance at every layer.

1. Energy Levels

$$E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}([SSq]), \quad n = 1, \dots, 26$$

where $E_0 = \hbar\omega_{\text{SCm}}$ and $\delta_n = (2\pi)^{n/6}$.

2. Element Mapping

n	Z_{id}	Element	Type
1	1	proto-H	Magnetic
14	14	proto-Si	Non-magnetic
26	26	proto-Fe	Magnetic

The ladder maps the complete magnetic/non-magnetic hierarchy from hydrogen through iron.

3. Source Data

- **File:** spectral_ladder_26state.py
 - **Session:** 214
 - **CP4 Class:** SpectralLadder26StateCalc (#536)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
 2. PAPER_953 — Ramanujan-Accelerated S_{26} Convergence
 3. PAPER_956 — Spectral Ladder Phonon Mapping
 4. PAPER_959 — 26D Ramanujan Summation
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Cross-Links

Related Paper	Relationship
PAPER_953	Ramanujan acceleration of S_{26}
PAPER_956	Phonon mapping $E_n \rightarrow \omega_n \rightarrow Q_n$
PAPER_949	BCS gap uses ladder as phonon channels
PAPER_959	Full 26D Ramanujan summation engine

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
$[SSq]$	—	0.57	BH dynamics
ω_{SCm}	—	$2\pi \times 1.25 \text{ THz}$	Base energy E_0
E_0	$\hbar\omega_{\text{SCm}}$	$8.27 \times 10^{-22} \text{ J}$	Ladder base

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
26 energy levels	$E_n = E_0(2\pi)^{n/3} S_{26}$	Derived
Magnetic/non-magnetic hierarchy	Z mapping 1-26	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Spectral Ladder (26-State HRes Phonon Hierarchy)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{ladder}} = \sum_{n=1}^{26} \left[\frac{1}{2} \dot{q}_n^2 - \frac{1}{2} \omega_n^2 q_n^2 \right], \quad \omega_n = E_n/\hbar$$

§A.3 Euler-Lagrange Equation of Motion

$$\ddot{q}_n + \omega_n^2 q_n = 0, \quad n = 1, \dots, 26$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow \omega_{\text{SCm}} \rightarrow$ spectral ladder $E_n \rightarrow$ 26-shell hierarchy $\rightarrow$ element mapping

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$VDS(n) = \rho_{\text{SCm}} \cdot (2\pi)^{n/3}$ — density profile across ladder levels.

§B.2 DVP

Prime sequence $\{2, 3, 5, 7, 11, 13, 17, 19, 23\}$ maps to magnetic shell closures at levels 2, 8, 14, 20, 26.

§B.3 BSH

$BSH(n) = \tanh(n/26) \cdot S_{26}$ — saturation envelope over the 26 levels.

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
BSH layers	26	Confirmed
$[SSq]$	0.57	Confirmed